Decimal Operations

All four operations, and a point that lands where it should.

THE BIG QUESTION

Why does multiplying by 0.5 make a number smaller, when multiplying is supposed to make things bigger?

HANDS-ON PROJECT

Unit Price Investigation

Real prices, real sizes. Find one product where the bigger box is the worse deal, and prove it.

FLUENCY GAME

Point Placement

Given the digits, race to place the decimal point. Same digits, four different answers.

AXIOM · MISSION COMMANDER, NINE TOURS

“Multiply the digits and forget the point. Count the decimal places, then put it back. Two jobs, done separately, both easy.”

MISSION 04 · BEFORE WE LAUNCH

NAME ______________

DATE ______________

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Item F2 asks you to predict rather than calculate — guess, and say why.

STRAND A ADDING	A1 3.5 + 0.47	A2 12.6 + 4.85
STRAND B SUBTRACTING	B1 4 – 1.35	B2 10 – 0.07
STRAND C MULTIPLYING	C1 0.4 × 0.2	C2 2.5 × 1.4
STRAND D DIVIDING	D1 4.8 ÷ 2	D2 6 ÷ 0.5
STRAND E MONEY	E1 $7.20 for 8 — price each	E2 3 items at $4.99 — the total
STRAND F PREDICTING SIZE	F1 40 × 0.75 — bigger or smaller than 40?	F2 Is 6 ÷ 0.5 bigger or smaller than 6? Why?

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the five-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Adding	A1 A2	Week 1 · lessons 1.1 and 1.2	___ / 2	Both days. This is Year One material revisited.
B · Subtracting	B1 B2	Week 1 · lessons 1.3 and 1.4	___ / 2	Both days. B2 is the one that matters — borrowing across a decimal point.
C · Multiplying	C1 C2	Week 2 · lessons 2.1 and 2.2	___ / 2	Compress Week 2 to three days. Keep 2.4 — why it gets smaller is the Big Question.
D · Dividing	D1 D2	Week 3 · lessons 3.1 to 3.3	___ / 2	Day 3.1 only. D2 passing is rare and worth checking — ask him to explain it out loud.
E · Money	E1 E2	Week 4 · lessons 4.1 and 4.2	___ / 2	Day 4.1. Keep 4.2 — two-step money problems are where the errors live.
F · Predicting size	F1 F2	Week 2 · lesson 2.4	___ / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

This mission runs five weeks, not four, because decimal division genuinely takes longer than the rest. If you need to save time, compress Week 1 — adding and subtracting decimals is revision. Do not compress Week 3.

NAME ___________

DATE ___________

Line Up the Point

Not the last digit — the point. Fill the gaps with zeros.

Warm-Up
SAME NUMBER OF PLACES

0.3 + 0.4	1.2 + 2.5	0.25 + 0.25	2.5 + 1.5	0.6 + 0.9	3.4 + 1.2

3.5 + 0.47 · THE ZERO THAT MAKES IT WORK

WRONG	RIGHT
3.5	3.50
+0.47	+0.47
3.52	3.97

The wrong version lined up the last digits, so the 7 hundredths got added to the 5 tenths. Writing 3.5 as 3.50 makes both numbers two places long and the columns line up on their own.

A trailing zero adds nothing to a number's value, which is exactly why you are allowed to add as many as you need.

Core
WRITE IN THE ZEROS FIRST

1. 3.5 + 0.47
2. 12.6 + 4.85
3. 9.4 + 0.68
4. 7.05 + 2.9
5. 0.125 + 0.875. Estimate first — the answer is a whole number, and knowing that in advance is the point.

Challenge
NOT GRADED

C1. 2.5 + 3.75 + 0.125. Three numbers, three different lengths. Line up all three at once.

C2. What adds to 4.6 to make 10? Do it without writing the subtraction down.

AXIOM: Make every number the same length before you start. Ten seconds writing zeros saves you the whole problem.

Subtracting Decimals

NAME ___________

DATE ___________

Same alignment. Borrow across the point when you must.

Warm-Up

0.9 - 0.4	3.7 - 1.2	1 - 0.5	2.5 - 0.5	0.75 - 0.25	5.8 - 2.3

4 - 1.35 · A WHOLE NUMBER IS A DECIMAL IN DISGUISE

```
  4.00
 -1.35
 -----
  2.65
```

Write 4 as 4.00 and the subtraction becomes ordinary. Without those zeros there is nothing in the hundredths column to take 5 from.

Check it by adding back: 2.65 + 1.35 = 4.00. Subtraction always gives you that check for free, and it takes five seconds.

Core

1. 4 - 1.35

2. 10 - 0.07

3. 8.2 - 3.45

4. 12.3 - 4.75

5. A 2.5 m board has 0.85 m cut off. How much is left? Write the check underneath.

Challenge

C1. 12.4 – 7.856. Three places against one — write in the zeros before you borrow.

C2. Somebody worked out 10 – 0.07 and got 9.3. Which column did they get wrong?

NAME ___________

DATE ___________

Estimate to Check

Round to whole numbers first. The estimate catches a misplaced point instantly.

Warm-Up

ROUND TO THE NEAREST WHOLE

4.7	3.2	9.8	0.4	12.5	7.05
____	____	____	____	____	____

THE ESTIMATE CATCHES THE POINT

12.4 - 7.856

About 12 – 8 = **4**.
An answer of 45 or 0.45 is wrong on sight.

2.5 × 1.4

About 3 × 1 = **3**.
So 3.5 is believable; 35 is not.

Why it works

A misplaced decimal point is always wrong by a factor of ten. That is the easiest kind of error to spot.

Core

ESTIMATE, COMPUTE, COMPARE

1. 9.8 - 4.9

2. 12.4 - 7.856

3. 0.49 × 100

4. 19.8 ÷ 5

5. Six items at $4.95. Estimate the total, then work it out exactly. How far off was the estimate?

Challenge

C1. Estimate 1.02 × 500 in your head. Why is this one easier than it looks?

C2. Which is a better estimate for 4.9 × 21 — 100 or 105? Both are reasonable, so say what you traded away.

NAME ___________

DATE ___________

Money Is Decimals

Two places, always. A dollar sign changes nothing about the arithmetic.

Warm-Up

CHANGE FROM A TWENTY

$20 - $5	$20 - $10.50	$10 - $2.50	$5 - $1.25	$1 - $0.35	$20 - $15
______	______	______	______	______	______

UNIT PRICE · THE ONLY FAIR COMPARISON

PACK	PRICE	EACH
4 for	$5.00	$1.25
8 for	$9.00	$1.125

Nine dollars is more than five, and eight is more than four. Neither number tells you which is the better deal — only the price each does.

Notice the answer has three decimal places. Money usually stops at two, but a unit price does not have to: $1.125 each is a real, useful number even though you cannot hand somebody half a cent.

Core

1. $7.20 for 8 — price each

2. $4.50 for 3 — price each

3. 3 items at $4.99 — the total

4. Change from $20 on that total

5. $5 for 400 g, or $9 for 800 g. Which is better value? Show the price per 100 g for both.

Challenge

C1. 1 kg costs $8, or 750 g costs $5.40. Work out the price per kilogram for the small one, then say which wins.

C2. Buy 2 get 1 free at $6 each. What is the real price each across all three?

NAME ___________
DATE ___________

Point Placement

Same digits, four different answers. Only the point moves.

How to play: one player writes a multiplication of two decimals. The other multiplies the digits with the points ignored, then races to place the point by counting decimal places. Say the count out loud — “two places in, two places out” — before writing the answer.

THE DIGITS 24, FOUR WAYS

6 × 4	6 × 0.4	0.6 × 0.4	0.06 × 0.4
24	2.4	0.24	0.024

Every one of these is 6 × 4 = 24. Count the decimal places going in, and put the same number in the answer.

Play six rounds

ROUND	PROBLEM	DIGITS ONLY	PLACES	ANSWER
1				
2				
3				
4				
5				
6				

Challenge

C1. The digits come out as 240 and the answer needs three decimal places. Write it. Then explain what happened to the last zero.

C2. Find three different multiplications that all give exactly 0.24.

Weeks 2–5

Every day below has five graded sets waiting in Practice Bay.

QUIZ · end of week 4
TEST · end of week 5

Week 2 · Multiplying Decimals

Multiply the digits, then count the places. Two separate jobs, both easy on their own.

MON 2.1	TUE 2.2	WED 2.3	THU 2.4	FRI
Ignore the point Multiply the digits first.	**Count the places** Two in, two out.	**Place it by estimating** 0.4 × 60 is about 24, so the point has one home.	**Smaller than you started** Why × 0.5 halves. The Big Question.	**Point Placement** Page 8's game, played again for speed.

Week 3 · Dividing Decimals HARDEST WEEK

Shift both numbers the same number of places, then divide as usual. Do not compress this week.

MON 3.1	TUE 3.2	WED 3.3	THU 3.4	FRI
Decimal ÷ whole The point comes straight up.	**Whole ÷ decimal** Shift both, then divide.	**Decimal ÷ decimal** Same shift, applied to both.	**Money problems** Unit prices and change, to the cent.	**Investigation begins** Collect real prices and sizes. No conclusions yet.

Week 4 · All Four Together QUIZ FRIDAY

Now the words have to tell him which operation to reach for.

MON 4.1	TUE 4.2	WED 4.3	THU 4.4	FRI
Choose the operation Read the words twice.	**Two-step problems** Multiply then subtract, in that order.	**Measurement** Metres, litres, kilograms.	**Estimate everything** Estimate above the working, every time.	**Mid-unit quiz** Page 10. Eight items, 85%.

Week 5 · Proof UNIT TEST

The investigation gets written up as a recommendation somebody could act on.

MON 5.1	TUE 5.2	WED 5.3	THU	FRI
Compute unit prices Every item on the list.	**Find the trap** One product where bigger costs more per unit.	**Write it up** One paragraph a shopper could use.	**Error journal sweep** Fix only what repeats.	**Mission 04 test** Pages 11–12.

Mid-Unit Quiz

NAME ______________

SCORE ______ / 8

Eight items from Weeks 1 to 4. 85% to keep flying — that's 7 out of 8.

1. 3.5 + 0.47 ....

2. 8.2 – 3.45 ....

3. 0.4 × 0.2 ....

4. 3.2 × 2.5 ....

5. 6 ÷ 0.5 ....

6. 9.6 ÷ 1.2 ....

7. $7.20 for 0.8 kg — the price per kg ....

8. 40 × 0.75 comes out smaller than 40. Explain why, using the word “tenths” or “parts” in your answer.

....

....

Marking note: item 8 wants the idea, not the product. Accept any answer saying that 0.75 is less than one whole, so you are taking part of 40 rather than 40 several times over.

ANSWERS ON PAGE 15

Unit Test · Part 1

NAME ______________

DATE ______________

Items 1–8. No time limit. Write an estimate above every calculation.

1. $1.2 + 2.5$ ....

2. $12.6 + 4.85$ ....

3. $4 - 1.35$ ....

4. $12.4 - 7.856$ ....

5. 1.2×0.5 ....

6. 1.25×0.8 ....

7. $7.35 \div 15$ ....

8. $7.5 \div 0.25$ ....

CONTINUES ON PAGE 12

Unit Test · Part 2

SCORE ______ / 12

BAND ______________

Items 9–12 and the Big Question.

9. \$4.50 for 3 kg — the price per kg ....

10. 3 items at \$4.99 — the change from \$20 ....

11. A 4.5 m rope is cut into 0.75 m pieces. How many pieces? Show why division is the right operation.

....

12. 400 g costs \$5, and 800 g costs \$9. Find the price per 100 g for each, and say which is better value.

....

EXPLAIN YOUR THINKING · THE BIG QUESTION

Why does multiplying by 0.5 make a number smaller, when multiplying is supposed to make things bigger?

Use one number as your example, and say what happens when you multiply by 1.25 instead. Scored on reasoning, not neatness.

....

....

....

....

....

11–12
Gold band. The Precision Badge is earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 3.97 | A2 · 17.45 | B1 · 2.65 | B2 · 9.93

C1 · 0.08 | C2 · 3.5 | D1 · 2.4 | D2 · 12

E1 · $0.90 | E2 · $14.97 | F1 · smaller | F2 · bigger

D2 and F2 are the diagnostic pair. Getting 12 without being able to say why it is bigger than 6 means the procedure is memorised and the meaning is not — teach Week 3 in full.

Lesson 1.1 · Line Up the Point

Warm-Up · 0.7 | 3.7 | 0.5 | 4 | 1.5 | 4.6

Core · 1 · 3.97 | 2 · 17.45 | 3 · 10.08 | 4 · 9.95 | 5 · 1

C1 · 6.375. C2 · 5.4. Both worth watching for method: writing all three numbers to three places before adding is the habit being built.

Lesson 1.2 · Subtracting Decimals

Warm-Up · 0.5 | 2.5 | 0.5 | 2 | 0.5 | 3.5

Core · 1 · 2.65 | 2 · 9.93 | 3 · 4.75 | 4 · 7.55 | 5 · 1.65 m

C1 · 4.544. C2 · they treated 0.07 as 0.7 and subtracted from the tenths column instead of the hundredths.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · Estimate to Check

Warm-Up · 5 | 3 | 10 | 0 | 13 | 7

Core · 1 · 4.9 | 2 · 4.544 | 3 · 49 | 4 · 3.96 | 5 · est $30, actual $29.70

C1 · about 500, because 1.02 is barely more than 1 — the true value is 510. C2 · 105 is closer but 100 is faster; the trade is accuracy against speed, and for catching a misplaced point either works.

Lesson 1.4 · Money Is Decimals

Warm-Up · $15 | $9.50 | $7.50 | $3.75 | $0.65 | $5

Core · 1 · $0.90 | 2 · $1.50 | 3 · $14.97 | 4 · $5.03

Item 5 · $1.25 per 100 g against $1.125 per 100 g, so the 800 g pack wins. C1 · $7.20 per kg for the 750 g, so it beats the $8 kilogram. C2 · $4 each.

Friday · Point Placement

C1 · 0.240, which is just 0.24 — the trailing zero holds no value, and noticing that is the whole lesson. C2 · many answers: 0.6 × 0.4, 6 × 0.04, 1.2 × 0.2, 0.8 × 0.3. Ask for three genuinely different ones rather than three rewrites of the same pair.

TEACHER PAGE · ANSWER KEY

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 3.97

2 · 4.75

3 · 0.08

4 · 8

5 · 12

6 · 8

7 · $9.00

8 · reasoning

Unit test

1 · 3.7

2 · 17.45

3 · 2.65

4 · 4.544

5 · 0.6

6 · 1

7 · 0.49

8 · 30

9 · $1.50

10 · $5.03

Items 11, 12 and the Big Question

11 · 6 pieces; division is right because the question asks how many 0.75s fit inside 4.5. **12** · $1.25 per 100 g against $1.125 per 100 g, so the 800 g is better value. **Big Question** · full marks for an answer explaining that 0.5 is half of one whole, so multiplying by it takes half of the number rather than adding copies of it — and that any factor below 1 shrinks while any factor above 1 grows. A student who says “multiplying always makes things bigger, so the rule is broken” has understood the question and not the answer: that is a conversation, not a mark.

Error journal · Mission 04

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 5, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY